

# 10.7.21

AI25BTECH11012 - GARIGE UNNATHI

**Question:**

Find the points on the curve  $x^2 + y^2 - 2x - 3 = 0$  at which the tangents are parallel to the x-axis.

**Solution:**

For the given curve

$$\mathbf{u} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \quad (0.1)$$

$$f = -3 \quad (0.2)$$

Hence the centre and radius are given as :

$$\mathbf{c} = -\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (0.3)$$

$$r = \sqrt{\|\mathbf{u}\|^2 - f} = 2 \quad (0.4)$$

For a circle the points of contact are given by the formula :

$$\mathbf{q}_{ij} = \left( \pm r \frac{\mathbf{n}_j}{\|\mathbf{n}_j\|} - \mathbf{u} \right) \quad (0.5)$$

Where  $i, j = 1, 2$

Since tangents are parallel to x - axis ,the normal is given as

$$\mathbf{n} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (0.6)$$

Solving 0.5 we get :

$$\mathbf{q} = (\pm 2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{pmatrix} -1 \\ 0 \end{pmatrix}) \quad (0.7)$$

$$= \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad (0.8)$$

Hence the two points of contact are

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad (0.9)$$

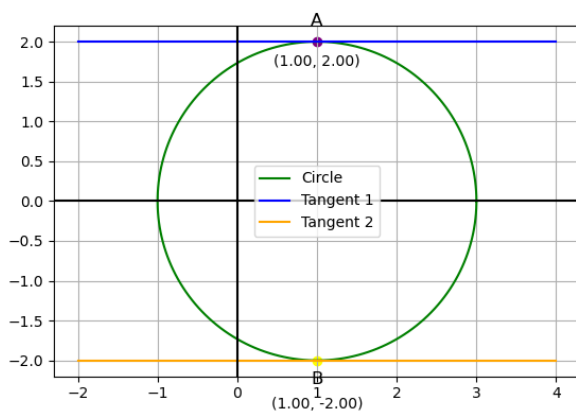


Fig. 0.1